

Customer Relationship Management System

Functional Workflow and Project Documentation

1. Project Overview

The Customer Relationship Management system manages the complete customer lifecycle, starting from lead collection and continuing through sales, customer onboarding, service delivery, support, renewals, and customer retention.

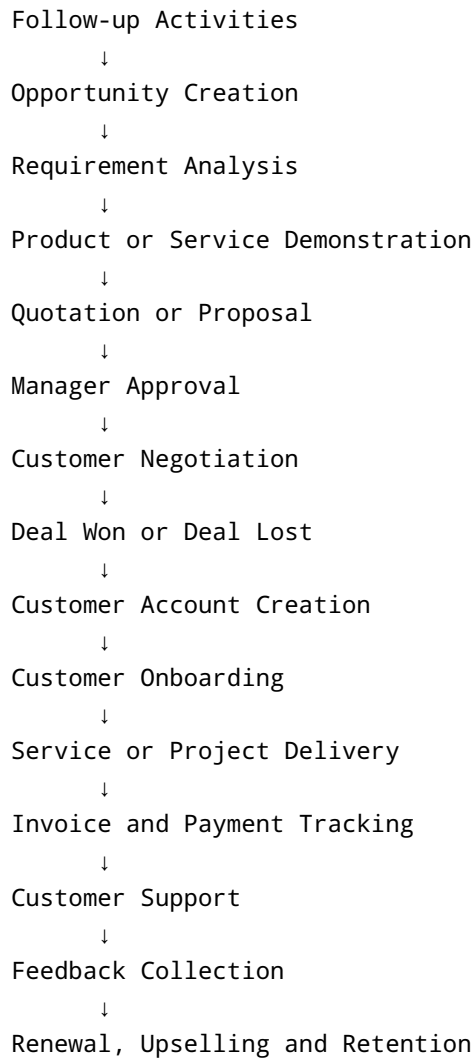
The CRM system provides a centralized place to manage:

- Leads
- Customers and companies
- Contacts
- Follow-ups
- Sales opportunities
- Quotations
- Activities and tasks
- Customer onboarding
- Support tickets
- Renewals
- Sales reports
- User roles and permissions

The primary goal is to ensure that no lead, customer request, follow-up, sales opportunity, or support issue is missed.

2. Complete CRM Workflow

```
Lead Generation
  ↓
Lead Capture
  ↓
Duplicate Verification
  ↓
Lead Assignment
  ↓
Initial Contact
  ↓
Lead Qualification
  ↓
```



3. CRM User Roles

3.1 Super Admin

The Super Admin manages:

- System configuration
- Companies or branches
- Users
- Roles
- Permissions
- CRM settings
- Email configuration
- Notification configuration

- Data backup
- Audit logs

3.2 CRM Manager

The CRM Manager manages:

- All leads
- Lead assignments
- Sales pipelines
- Team performance
- Approval requests
- Lead reassignment
- Lost-deal analysis
- Reports and dashboards

3.3 Sales Manager

The Sales Manager manages:

- Sales executives
- Opportunities
- Quotation approvals
- Discounts
- Sales targets
- Pipeline forecasting
- Deal reviews

3.4 Sales Executive

The Sales Executive manages:

- Assigned leads
- Calls and meetings
- Follow-ups
- Opportunities
- Quotations
- Customer communication
- Sales activities
- Deal closure

3.5 Marketing Executive

The Marketing Executive manages:

- Campaigns

- Lead sources
- Website leads
- Social media leads
- Email campaigns
- Campaign performance
- Lead generation reports

3.6 Support Executive

The Support Executive manages:

- Customer complaints
- Support tickets
- Ticket assignment
- Resolution updates
- Customer communication
- Service-level agreement tracking

3.7 Accounts User

The Accounts User manages:

- Customer invoices
- Payment status
- Outstanding balances
- Payment reminders
- Receipts
- Refund details

3.8 Management or CEO

Management can view:

- CRM dashboard
- Sales performance
- Revenue forecast
- Conversion rate
- Employee performance
- Won and lost deals
- Customer retention
- Outstanding payments

4. CRM Dashboard

The CRM dashboard should show the most important business information.

Dashboard KPI Cards

- Total Leads
- New Leads
- Contacted Leads
- Qualified Leads
- Unqualified Leads
- Active Opportunities
- Won Deals
- Lost Deals
- Expected Revenue
- Confirmed Revenue
- Follow-ups Due Today
- Overdue Follow-ups
- Open Support Tickets
- Renewals Due

Dashboard Charts

- Lead Source Distribution
- Leads by Status
- Sales Pipeline
- Monthly Revenue
- Won versus Lost Deals
- Lead Conversion Trend
- Sales Executive Performance
- Customer Acquisition Trend
- Ticket Status Distribution
- Renewal Forecast

Dashboard Quick Actions

- Add Lead
 - Add Customer
 - Schedule Follow-up
 - Create Opportunity
 - Create Quotation
 - Add Task
 - Schedule Meeting
 - Create Support Ticket
-

5. Lead Management

5.1 Lead Sources

Leads may come from:

- Website
- Phone call
- Email
- Social media
- Advertisement
- Referral
- Walk-in
- Trade show
- Campaign
- Existing customer
- Partner
- Manual entry
- Imported file
- API integration

5.2 Lead Creation Fields

Each lead should contain:

- Lead ID
- Lead name
- Company name
- Mobile number
- Alternative mobile number
- Email address
- Lead source
- Interested product or service
- Industry
- Location
- Expected budget
- Priority
- Assigned employee
- Lead status
- Next follow-up date
- Description
- Created date
- Created by

5.3 Lead Statuses

Recommended lead statuses:

```
New
Assigned
Contact Attempted
Contacted
Follow-up Required
Qualified
Unqualified
Converted
Lost
Duplicate
On Hold
```

5.4 Lead Assignment Workflow

```
New Lead Created
↓
System Checks Duplicate Lead
↓
CRM Manager Reviews Lead
↓
Lead Assigned to Sales Executive
↓
Sales Executive Receives Notification
↓
Sales Executive Contacts Lead
↓
Lead Status Updated
```

Lead assignment methods:

- Manual assignment
- Round-robin assignment
- Location-based assignment
- Product-based assignment
- Department-based assignment
- Employee workload-based assignment

5.5 Duplicate Lead Validation

The system should check:

- Mobile number
- Email address
- Company name
- Tax registration number
- Existing customer records

When a duplicate is identified, the system should allow the user to:

- View the existing record
 - Merge the records
 - Cancel the new record
 - Continue with manager permission
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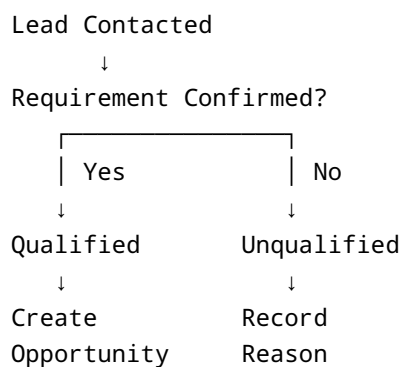
6. Lead Qualification

Lead qualification determines whether the lead is suitable for conversion into a sales opportunity.

Qualification Questions

- Does the customer have a genuine requirement?
- Does the customer have an approved budget?
- Is the decision-maker identified?
- Is there an expected purchase date?
- Is the requested service available?
- Is the customer interested in a demonstration?
- Is the opportunity commercially viable?

Qualification Result



Unqualified Reasons

- No requirement
 - No budget
 - Invalid contact
 - Service unavailable
 - Outside service area
 - Student or research enquiry
 - Duplicate enquiry
 - Competitor selected
 - Customer not interested
 - Unable to contact
-

7. Follow-up Management

Every customer interaction should be recorded as a follow-up activity.

Follow-up Types

- Phone call
- Email
- WhatsApp
- Meeting
- Video meeting
- Product demonstration
- Site visit
- Proposal discussion
- Payment follow-up
- Renewal follow-up

Follow-up Fields

- Lead or customer
- Assigned employee
- Follow-up type
- Scheduled date
- Scheduled time
- Discussion notes
- Customer response
- Follow-up outcome
- Next action
- Next follow-up date
- Reminder time
- Status

Follow-up Statuses

Scheduled
Completed
Rescheduled
Missed
Cancelled
Overdue

Follow-up Workflow

```
graph TD
    A[Follow-up Scheduled] --> B[Reminder Sent]
    B --> C[Sales Executive Contacts Customer]
    C --> D[Discussion Recorded]
    D --> E[Outcome Selected]
    E --> F[Next Follow-up Scheduled or Lead Closed]
```

The system should automatically mark a follow-up as overdue when the scheduled time has passed without completion.

8. Opportunity Management

A qualified lead should be converted into an opportunity.

Opportunity Fields

- Opportunity ID
- Opportunity name
- Customer or company
- Contact person
- Product or service
- Expected deal value
- Expected closing date
- Probability percentage
- Sales stage
- Assigned employee

- Competitor details
- Customer requirements
- Decision-maker
- Next action
- Remarks

Sales Opportunity Stages

Requirement Identified
 Initial Discussion
 Qualified Opportunity
 Demo Scheduled
 Demo Completed
 Proposal Preparation
 Proposal Sent
 Negotiation
 Awaiting Customer Approval
 Awaiting Internal Approval
 Won
 Lost
 On Hold

Opportunity Workflow

```

    Qualified Lead
      ↓
    Create Opportunity
      ↓
    Capture Customer Requirement
      ↓
    Estimate Deal Value
      ↓
    Schedule Demonstration
      ↓
    Prepare Proposal
      ↓
    Send Proposal
      ↓
    Negotiate
      ↓
    Won, Lost or On Hold
  
```

9. Product or Service Demonstration

The CRM should allow users to schedule and record demonstrations.

Demo Fields

- Customer
- Opportunity
- Demo date
- Demo time
- Demo type
- Demo location or online meeting link
- Presented by
- Participants
- Products or services demonstrated
- Customer feedback
- Demo result
- Next action

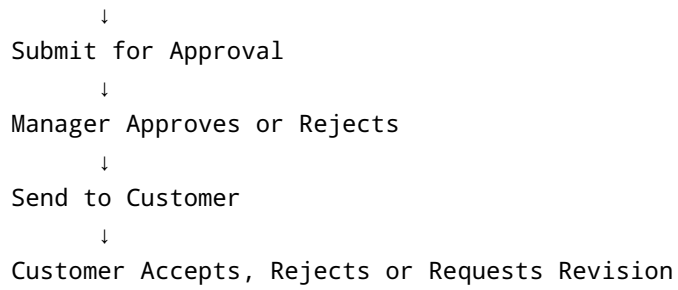
Demo Statuses

Scheduled
Confirmed
Completed
Rescheduled
Cancelled
No Show

10. Quotation and Proposal Management

Quotation Workflow

Opportunity Selected
↓
Create Quotation
↓
Add Products or Services
↓
Calculate Price, Tax and Discount
↓
Add Terms and Conditions



Quotation Fields

- Quotation number
- Customer
- Contact person
- Opportunity
- Quotation date
- Expiry date
- Products or services
- Quantity
- Unit price
- Discount
- Tax
- Subtotal
- Grand total
- Payment terms
- Delivery terms
- Validity
- Notes
- Approval status

Quotation Statuses

Draft
Pending Approval
Approved
Rejected
Sent
Viewed
Revision Requested
Revised
Accepted
Declined
Expired
Converted

Approval Rules

Approval may be required when:

- Discount exceeds the permitted limit
 - Deal value exceeds the employee's authority
 - Payment terms are unusual
 - Margin is below the required limit
 - Special terms are added
 - Free services are included
-

11. Negotiation and Deal Closure

During negotiation, the system should record:

- Customer objections
- Requested discount
- Revised price
- Requested delivery date
- Payment conditions
- Competitor offer
- Final decision date
- Management approval

Deal Won Workflow

```
graph TD; A[Customer Accepts Proposal] --> B[Mark Opportunity as Won]; B --> C[Convert Lead into Customer]; C --> D[Create Customer Account]; D --> E[Create Agreement or Sales Order]; E --> F[Start Customer Onboarding];
```

Deal Lost Workflow

```
graph TD; A[Customer Rejects Proposal] --> B[ ];
```

Mark Opportunity as Lost

↓

Select Lost Reason

↓

Record Competitor

↓

Add Final Remarks

↓

Close Opportunity

Lost Deal Reasons

- Price too high
- Competitor selected
- No budget
- Requirement cancelled
- Project postponed
- No customer response
- Product unavailable
- Management rejected
- Payment terms not accepted
- Customer chose internal solution

12. Customer and Account Management

After a deal is won, the lead should be converted into a customer.

Customer Information

- Customer ID
- Customer name
- Company name
- Customer type
- Industry
- Phone
- Email
- Website
- Billing address
- Service address
- Tax number
- Account manager
- Customer status
- Credit limit
- Payment terms

- Registration date

Customer Profile Tabs

The customer profile should contain:

- Overview
- Contacts
- Opportunities
- Quotations
- Agreements
- Activities
- Tasks
- Meetings
- Documents
- Invoices
- Payments
- Support tickets
- Projects
- Renewals
- Notes
- Audit history

Customer Statuses

Prospective
Active
Inactive
On Hold
Blocked
Former Customer

13. Contact Management

One company may have multiple contact persons.

Contact Fields

- Contact name
- Company
- Job title
- Department
- Phone number

- Email address
- Decision-maker status
- Primary contact status
- Preferred communication method
- Notes

Contact types may include:

- Decision-maker
- Technical contact
- Billing contact
- Project contact
- Support contact
- General contact

14. Activity Management

The CRM should maintain a complete timeline of customer interactions.

Activity Types

- Call
- Email
- WhatsApp message
- Meeting
- Demo
- Site visit
- Follow-up
- Note
- Task
- Quotation
- Support ticket
- Payment reminder

Each activity should record:

- Date and time
 - Activity type
 - Employee
 - Customer or lead
 - Description
 - Outcome
 - Attachment
 - Next action
-

15. Task Management

Task Fields

- Task title
- Description
- Related lead, customer or opportunity
- Assigned employee
- Priority
- Start date
- Due date
- Reminder
- Status
- Completion notes

Task Statuses

Open
In Progress
Waiting
Completed
Cancelled
Overdue

Task Priorities

Low
Medium
High
Urgent

16. Calendar and Meeting Management

The CRM calendar should display:

- Follow-ups
- Calls
- Meetings
- Demos
- Tasks
- Quotation expiry dates

- Payment due dates
- Contract renewal dates

Meeting fields:

- Meeting title
- Customer
- Participants
- Date
- Start time
- End time
- Location
- Meeting link
- Agenda
- Minutes of meeting
- Action items
- Next meeting date

17. Customer Onboarding

Customer onboarding begins after the deal is won.

Onboarding Workflow

```
Deal Won
↓
Create Customer Account
↓
Collect Required Documents
↓
Confirm Scope and Deliverables
↓
Assign Account or Project Manager
↓
Create Implementation Plan
↓
Configure Product or Service
↓
Customer Training
↓
Customer Acceptance
↓
Go Live
```

↓
Onboarding Completed

Onboarding Checklist

- Agreement signed
- Customer details verified
- Required documents received
- Initial payment received
- Project owner assigned
- Scope confirmed
- Access credentials created
- Product configuration completed
- Training completed
- Go-live approved
- Handover completed

Onboarding Statuses

Not Started
Documents Pending
Payment Pending
In Progress
Training Scheduled
Ready for Go Live
Completed
On Hold
Cancelled

18. Support Ticket Management

Ticket Workflow

Customer Reports Issue
↓
Ticket Created
↓
Priority Assigned
↓
Support Executive Assigned
↓
Issue Investigated

↓
Resolution Provided
↓
Customer Confirmation
↓
Ticket Closed

Ticket Fields

- Ticket number
- Customer
- Contact person
- Subject
- Description
- Category
- Priority
- Assigned employee
- Created date
- Due date
- SLA target
- Resolution
- Status

Ticket Statuses

Open
Assigned
In Progress
Waiting for Customer
Waiting for Internal Team
Escalated
Resolved
Closed
Reopened

Ticket Priority

Low
Medium
High
Critical

19. Invoice and Payment Tracking

The CRM does not need to perform full accounting, but it should display customer billing information.

Payment Workflow

```
Deal Won
  ↓
Invoice Created
  ↓
Invoice Sent to Customer
  ↓
Payment Due Date Tracked
  ↓
Reminder Sent
  ↓
Payment Received
  ↓
Receipt Recorded
```

Invoice Statuses

```
Draft
Sent
Partially Paid
Paid
Overdue
Cancelled
Refunded
```

The CRM should show:

- Invoice number
 - Invoice amount
 - Amount received
 - Outstanding amount
 - Due date
 - Payment status
 - Payment method
 - Receipt reference
-

20. Renewal Management

The system should monitor customer contracts and subscriptions.

Renewal Workflow

```
graph TD; A[Contract Near Expiry] --> B[Automatic Renewal Reminder]; B --> C[Account Manager Contacts Customer]; C --> D[Create Renewal Opportunity]; D --> E[Prepare Renewal Quotation]; E --> F[Customer Negotiation]; F --> G[Renewed or Not Renewed];
```

Renewal Fields

- Customer
- Current contract
- Start date
- Expiry date
- Renewal value
- Renewal owner
- Reminder date
- Renewal probability
- Status
- Reason for non-renewal

Renewal Statuses

```
Upcoming
Contacted
Quotation Sent
Negotiation
Renewed
Not Renewed
Expired
```

21. Upselling and Cross-Selling

The CRM should help identify additional sales opportunities.

Examples:

- Upgrade existing plan
- Add more users
- Purchase additional products
- Add maintenance service
- Extend support
- Add training
- Add integrations
- Renew for a longer duration

Upselling opportunities should use the same opportunity and quotation workflow as new sales.

22. Notifications and Reminders

The system should send notifications for:

- New lead assigned
- Follow-up due
- Follow-up overdue
- Meeting reminder
- Task assigned
- Task overdue
- Quotation approval required
- Quotation accepted
- Deal won
- Deal lost
- Invoice overdue
- Support ticket assigned
- SLA breach warning
- Contract renewal approaching

Notification channels may include:

- In-app notification
 - Email
 - SMS
 - WhatsApp
 - Mobile push notification
-

23. CRM Reports

Lead Reports

- Lead source report
- Lead status report
- Lead assignment report
- Lead ageing report
- Uncontacted lead report
- Lead conversion report
- Lost lead report

Sales Reports

- Opportunity pipeline
- Won-deal report
- Lost-deal report
- Sales forecast
- Revenue report
- Product or service sales report
- Sales executive performance
- Target versus achievement
- Average deal size
- Average sales cycle

Activity Reports

- Calls completed
- Meetings completed
- Follow-ups completed
- Overdue follow-ups
- Employee activity report
- Customer interaction report

Customer Reports

- New customer report
- Active customer report
- Inactive customer report
- Customer revenue report
- Customer retention report
- Customer renewal report

Support Reports

- Open tickets

- Resolved tickets
- Overdue tickets
- SLA compliance
- Ticket category report
- Support executive performance

24. CRM KPIs

KPI	Description
Total Leads	Number of leads received
Contact Rate	Percentage of leads contacted
Qualification Rate	Percentage of leads qualified
Lead Conversion Rate	Leads converted into customers
Win Rate	Won opportunities compared with closed opportunities
Sales Pipeline Value	Total value of active opportunities
Expected Revenue	Opportunity value based on probability
Average Deal Value	Average value of won deals
Sales Cycle Length	Average time from lead creation to closure
Follow-up Completion	Completed follow-ups compared with scheduled follow-ups
Customer Retention	Customers continuing after the initial contract
Renewal Rate	Contracts successfully renewed
Ticket Resolution Time	Average support resolution duration
SLA Compliance	Tickets resolved within the agreed time

25. Suggested CRM Menu Structure

Dashboard

Leads

- └ All Leads
- └ My Leads
- └ Add Lead
- └ Lead Assignment

- └─ Lead Sources
- └─ Import Leads

Sales

- └─ Opportunities
- └─ Sales Pipeline
- └─ Demonstrations
- └─ Quotations
- └─ Approvals
- └─ Won Deals
- └─ Lost Deals

Customers

- └─ Customer List
- └─ Companies
- └─ Contacts
- └─ Customer Activities
- └─ Customer Documents

Activities

- └─ Follow-ups
- └─ Calls
- └─ Meetings
- └─ Tasks
- └─ Calendar
- └─ Reminders

Onboarding

- └─ New Onboarding
- └─ Active Onboarding
- └─ Checklist
- └─ Completed Onboarding

Support

- └─ Tickets
- └─ Ticket Categories
- └─ SLA
- └─ Escalations

Billing

- └─ Invoices
- └─ Payments
- └─ Outstanding Payments
- └─ Payment Follow-ups

Renewals

- └─ Upcoming Renewals
- └─ Renewal Opportunities
- └─ Renewed
- └─ Not Renewed

Reports

- └─ Lead Reports
- └─ Sales Reports
- └─ Activity Reports
- └─ Customer Reports
- └─ Support Reports
- └─ Performance Reports

Administration

- └─ Users
- └─ Roles
- └─ Permissions
- └─ Teams
- └─ Status Configuration
- └─ Email Templates
- └─ Notification Settings
- └─ Audit Logs

26. Recommended Database Entities

The CRM database should contain the following main entities:

- Users
- Roles
- Permissions
- Teams
- Leads
- Lead Sources
- Lead Assignments
- Lead Status History
- Companies
- Contacts
- Customers
- Opportunities
- Opportunity Stages
- Products
- Services
- Demonstrations
- Quotations

- Quotation Items
 - Quotation Approvals
 - Activities
 - Follow-ups
 - Tasks
 - Meetings
 - Notes
 - Documents
 - Onboarding Records
 - Onboarding Checklists
 - Support Tickets
 - Ticket Comments
 - Invoices
 - Payments
 - Contracts
 - Renewals
 - Notifications
 - Audit Logs
-

27. Important Business Rules

1. A new lead must be checked for duplicates before saving.
2. Every active lead must have an assigned employee.
3. A lead cannot be converted without being qualified.
4. A qualified lead can create one or more opportunities.
5. Every completed follow-up must contain an outcome.
6. A future follow-up should be scheduled when further contact is required.
7. High discounts must require manager approval.
8. A quotation cannot be sent before approval when approval rules apply.
9. A won deal must create or link to a customer account.
10. A lost deal must contain a lost reason.
11. Customer onboarding begins only after deal confirmation.
12. Ticket priority should determine its SLA target.

13. Closed tickets can only be changed by reopening them.
 14. Renewal reminders should be created before the contract expiry date.
 15. Deleted or modified important records must be recorded in audit logs.
-

28. Recommended Development Phases

Phase 1: Core CRM

- User management
- Roles and permissions
- Dashboard
- Leads
- Lead assignment
- Follow-ups
- Activities
- Customers
- Contacts

Phase 2: Sales Management

- Opportunities
- Sales pipeline
- Demonstrations
- Quotations
- Approval workflow
- Won and lost deals

Phase 3: Customer Operations

- Customer onboarding
- Tasks and calendar
- Documents
- Email notifications
- Support tickets

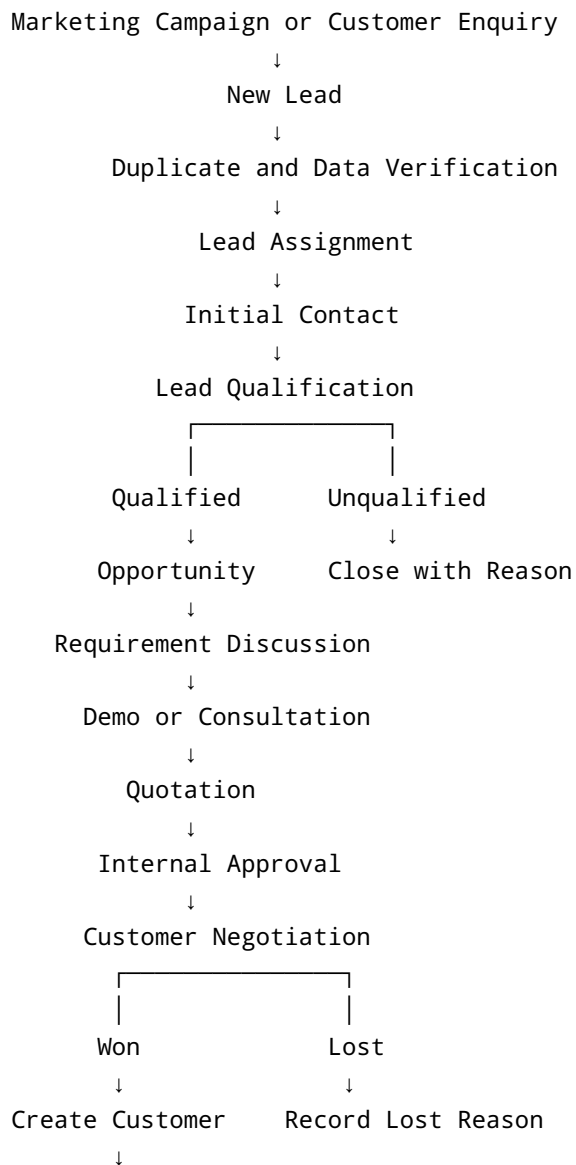
Phase 4: Commercial Management

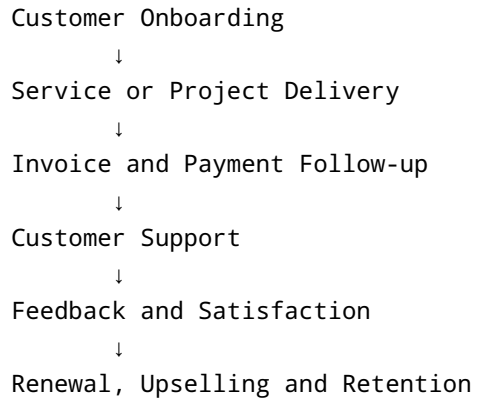
- Invoice tracking
- Payment follow-up
- Contracts
- Renewals
- Upselling

Phase 5: Reporting and Automation

- CRM reports
- Employee performance
- Automated lead assignment
- Follow-up reminders
- Workflow automation
- Audit logs
- Data export

29. Final End-to-End CRM Process





This workflow provides a complete CRM foundation for frontend development, backend APIs, database design, role permissions, testing, and project documentation.